Using Undergraduate GPA to Identify Combined Degree Candidates

Office of the Provost

2025-02-18

Question

The University would like to identify undergraduate students that would likely succeed as graduate students in combined degree programs. A GPA cut-off should be determined, above which undergraduate students will be invited to apply for a combined degree program. Additionally, we would like to know if GPA is more informative at a students Junior or Senior year.

Methodology:

Undergraduate students from the past 10 years were divided into three mutually exclusive categories:

- **Group 1: Undergraduate Stopouts** Students who started an undergraduate program and did not complete it at St. John's.
- **Group 2: Undergraduate Completers - No GR** Students who completed their undergraduate degrees but did not continue to a graduate program at St. John's
- **Group 3: Undergraduate Completers plus GR** Students who completed their undergraduate degrees and continued onto a graduate program at St. John's.

Students' cumulative GPAs were calculated at two separate points in time:

- 60 - 75 Credit Hours
- 75 - 90 Credit Hours

The following students were removed from the analysis:

- Transfer students who had a 0.0 undergraduate GPA at the time period under analysis.
- Students who are currently enrolled in an Undergraduate program as of Spring 2025.
- Students who completed their undergraduate degree but did not complete their graduate degree (ie, "Graduate Stopouts"). This category only accounts for <10% of students who complete their undergraduate degree and continue onto a graduate program.

GPA distributions of each group were tested for statistical significance using an ANOVA test at each point in time (figures 1.1a, 1.2a, 2.1a, 2.2a). Additionally, correlation between undergraduate GPA and cumulative graduate GPA was measured for group 3 to determine if undergraduate GPA would predict performance in a graduate program (figures 1.1b, 1.2b, 2.1b, 2.2b).

Finally, a graduate GPA of 3.25 was chosen to represent "success" in a graduate program. Various undergraduate GPA cutoffs were tested using ROC analysis to determine which cutoff was most useful in predicting success in a graduate program. The optimal cutoff was chosen by maximizing for Youden's J statistic (figures 3.1 and 3.2). Further analysis could be done using different graduate GPAs as the target for success.

Results:

For both the total student population and the Tobin subset, GPA differences between the three groups are statistically significant ($p \approx 0$) at both time periods. However, undergraduate GPA has a weak to moderate correlation with cumulative graduate GPA ($r <= 0.5$), and the correlation does not change drastically from the 60-75 credit range to the 75-90 credit range. Therefore, undergraduate GPA alone is only a moderately useful predictor of performance in a graduate program.

ROC analysis shows that undergraduate GPA cutoff of roughly 3.5 to be the most useful in predicting an ultimate graduate GPA >=3.25 ($f1 \approx 0.7$-0.8). This holds true for the total student population as well as the Tobin subset, and at both time periods.

Recommendations

- Given that undergraduate GPA does not become more informative from the 60-75 credit to the 75-90 credit range, there is no evident advantage to postponing invitations to combined degree programs.
- An undergraduate GPA of 3.5 is the optimal cutoff for predicting an eventual graduate GPA >= 3.25. However, the GPA cutoff for invitations could be lowered depending on the perceived risk of false positives (ie, inviting students to combined degree programs that ultimately earn a graduate GPA < 3.25).
- The impact that invitations to combined programs will have on graduate enrollment is unknown (ie, converting Group 2 students to Group 3). Therefore, follow-up analysis needs to be done once invitations have been sent out.

All Students: 2014-Present

60-75 Credits

Figure 1.1a: GPA Distribution by Group

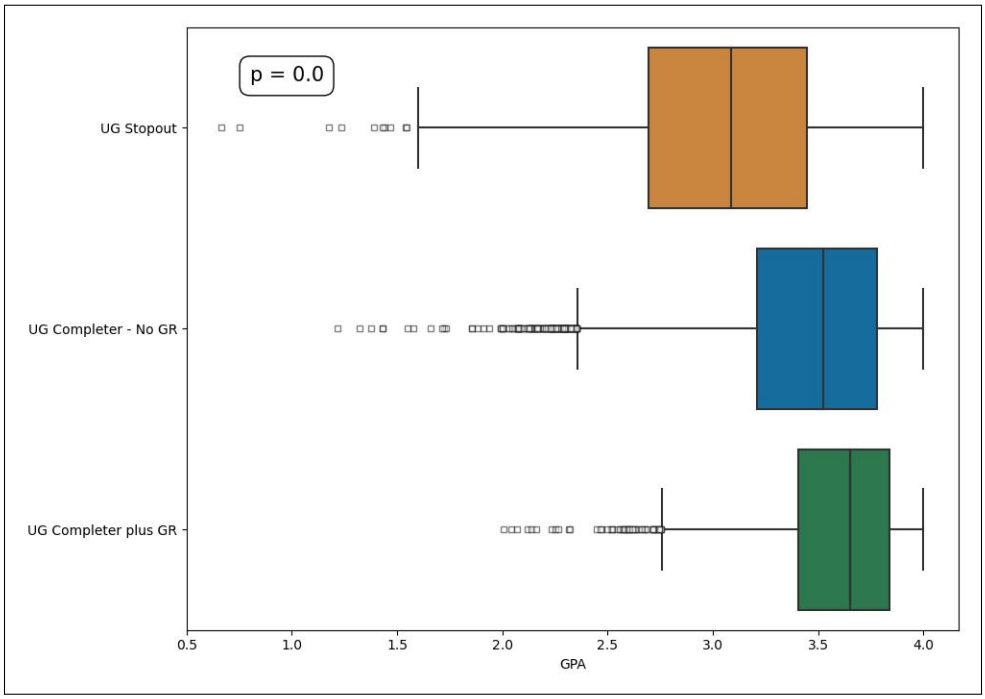
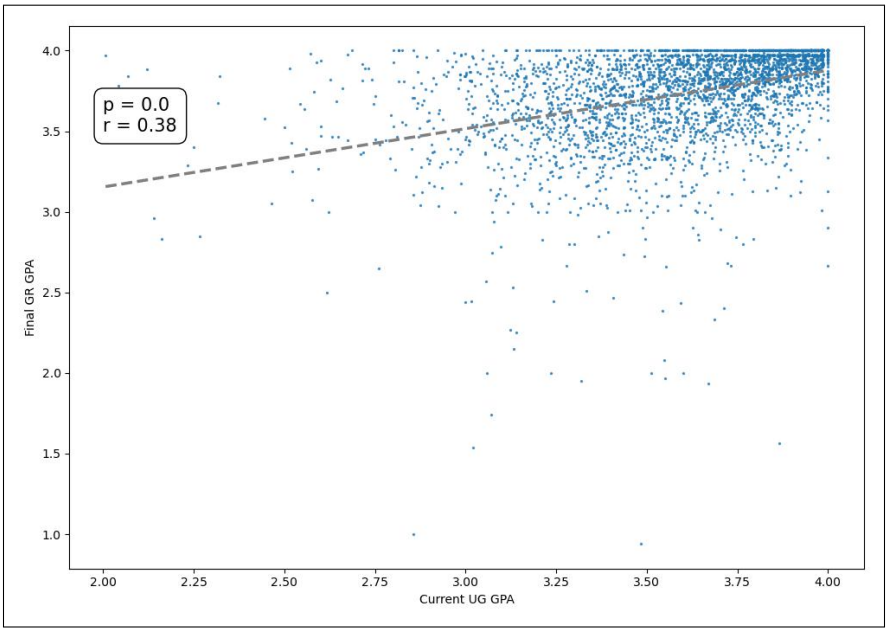


Figure 1.1b: Correlation of UG GPA and GR GPA of Group 3



75-90 Credits

Figure 1.2a: GPA Distribution by Group at

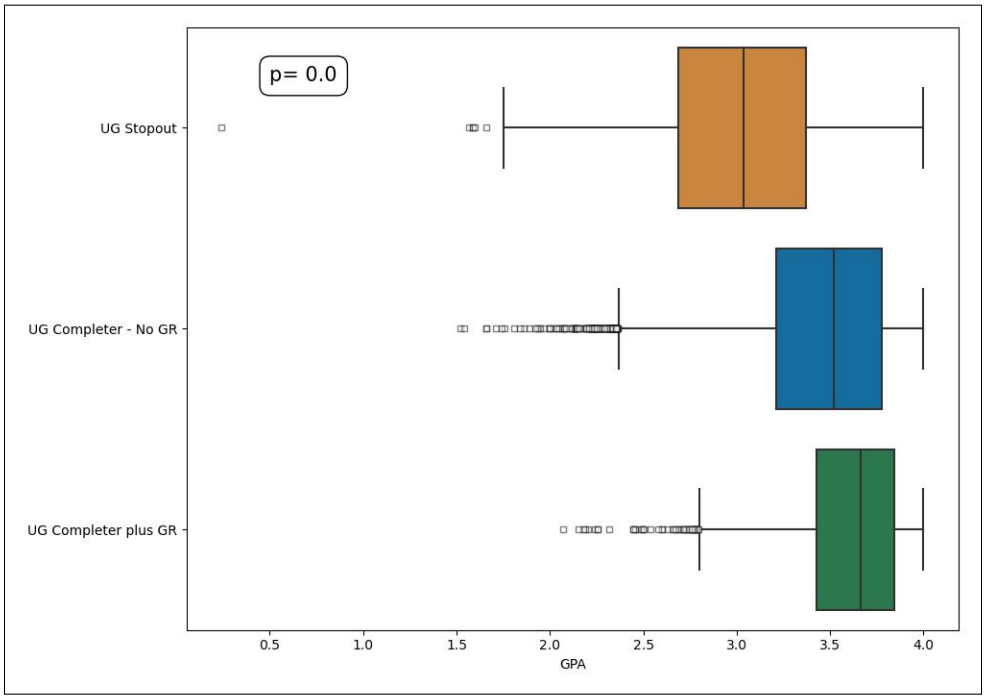
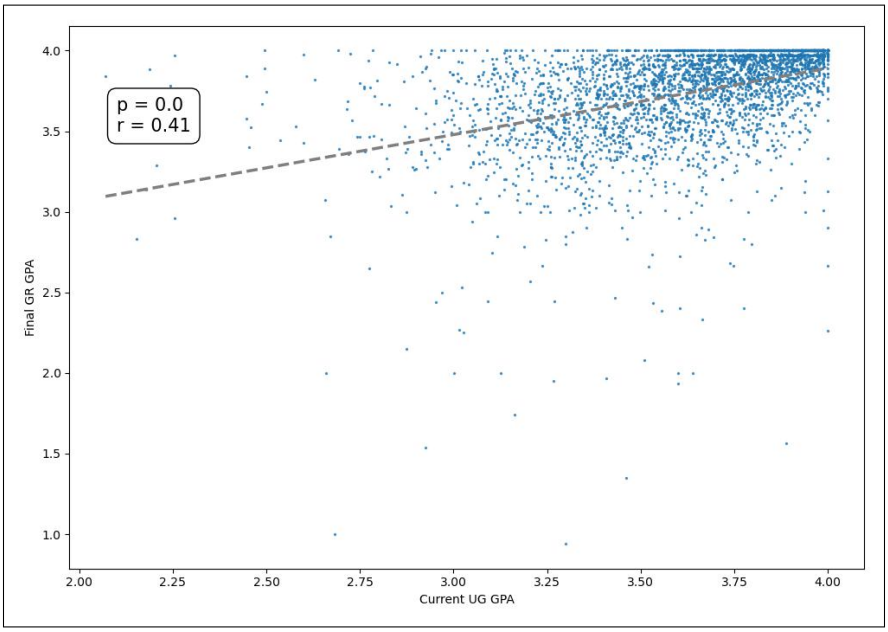


Figure 1.2b: Correlation of UG GPA and GR GPA of Group 3



Tobin Students: 2014-Present

60-75 Credits

Figure 2.1a: GPA Distribution by Group

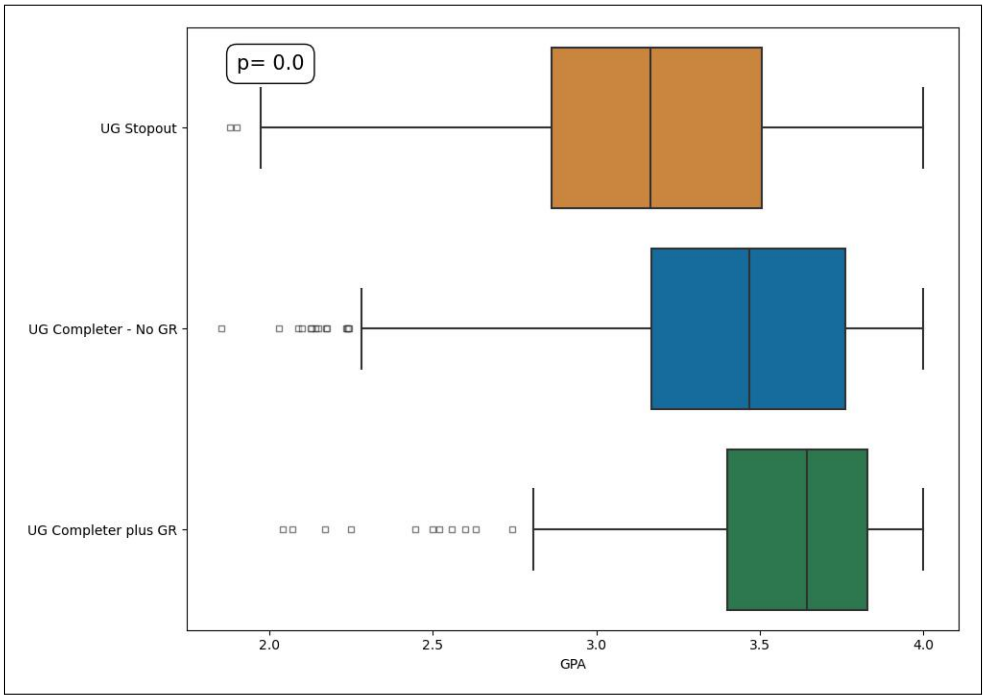
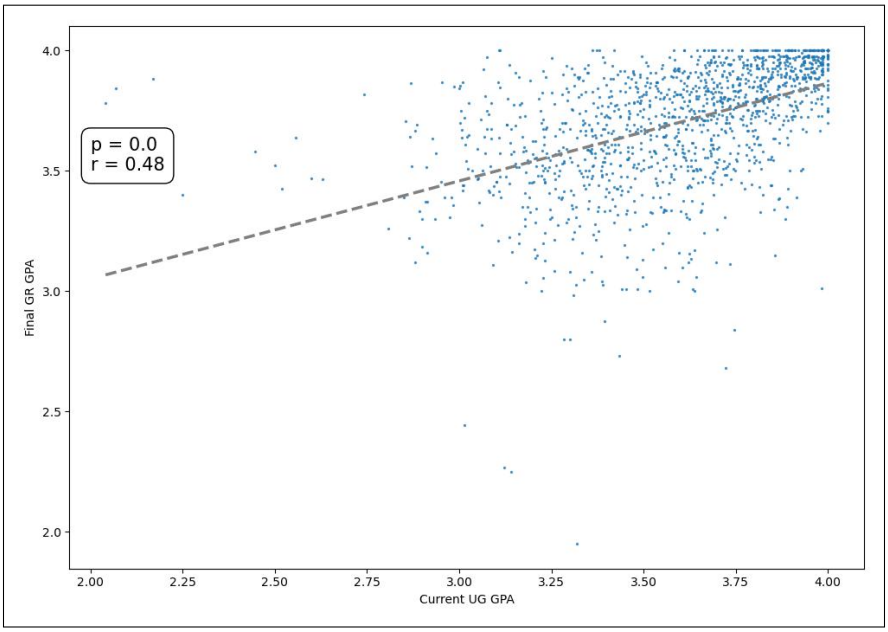


Figure 2.1b: Correlation of UG GPA and GR GPA of Group 3



75-90 Credits

Figure 2.2a: GPA Distribution by Group

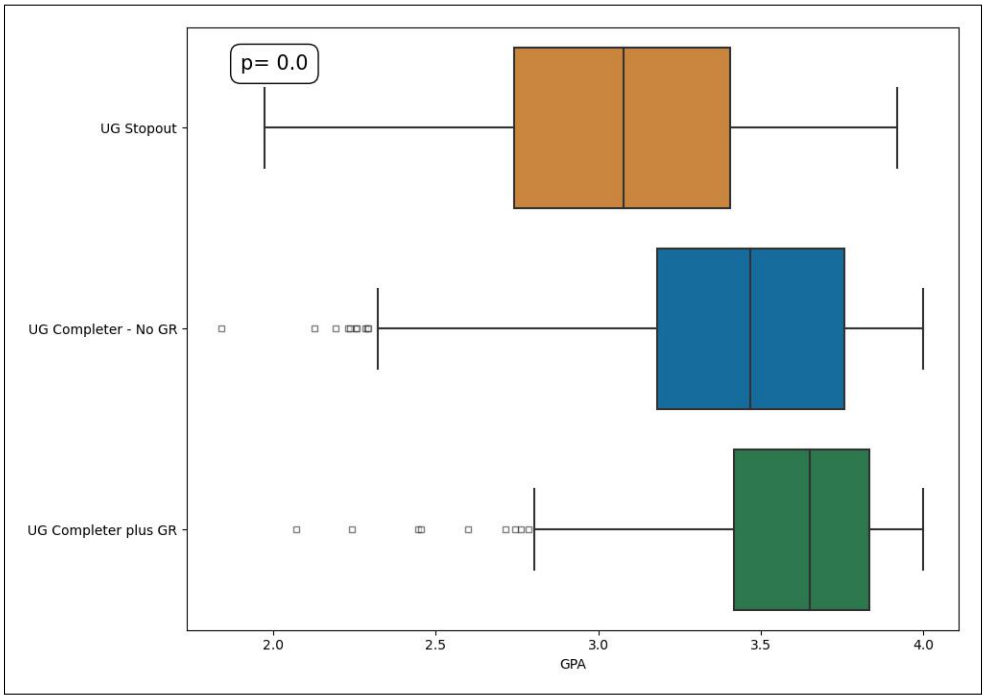
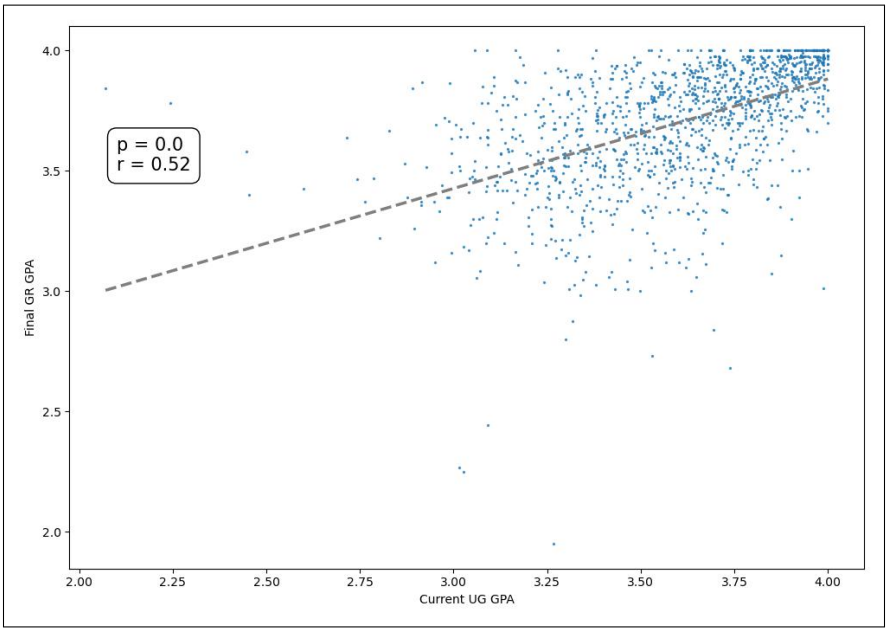
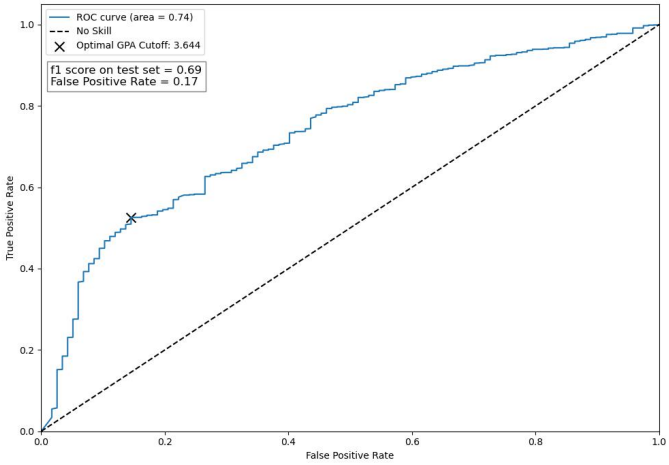


Figure 2.2b: Correlation of UG GPA and GR GPA of Group 3

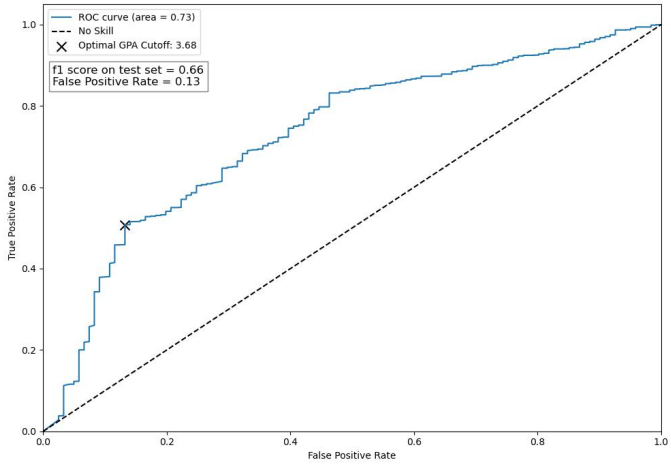


GPA Cutoff Testing and Validation

Figure 3.1: All Students

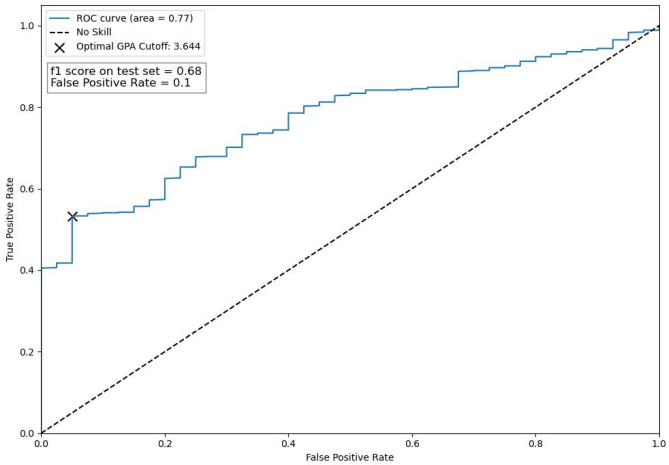


60-75 Credits

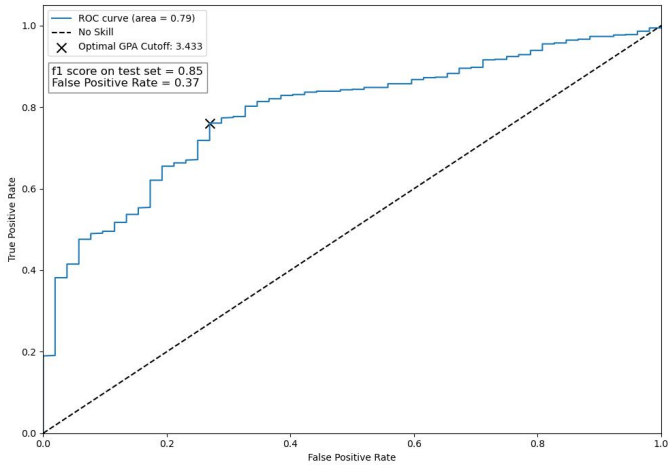


75-90 Credits

Figure 3.2 Tobin Students



60-75 Credits



75-90 Credits